E2 pin-to-pin connection table - DESIGN HOLD

All physical connections remain provisional or blocked. INTERNAL_VERIFY means inspect an existing module bond, not install another thin return. Component definitions, exact notes and release gates are in FINAL_WIRING.md.

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W001 / BLOCKED_RATING / WIRE	BT1.PACK+ -> F0.IN / PACK_RAW	power_hc / Pack: up to 12.6 V / Red	Main fuse adjacent to pack; insulated connector	Verify chemistry, polarity, pack C rating and connector interrupt/short-circuit capability
W002 / BLOCKED_RATING / WIRE	F0.OUT -> S0.COM / PACK_FUSED	power_hc / Pack voltage / Red	F0 protects feed	Select F0 from measured/spec start/stall currents and conductor ampacity
W003 / BLOCKED_MISSING / WIRE	S0.OUT -> P1.IN+ / PACK_ISOLATED	power_hc / Pack voltage / Red	Master isolation plus P1	Verify selected protection assembly input/output polarity and high-side topology
W004 / BLOCKED_MISSING / WIRE	P1.OUT+ -> TB_POS.BUS / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W005 / BLOCKED_MISSING / WIRE	TB_POS.BUS -> K1.MAIN_COM / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W006 / BLOCKED_MISSING / WIRE	TB_POS.BUS -> F1.IN / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W007 / BLOCKED_MISSING / WIRE	TB_POS.BUS -> F2.IN / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W008 / BLOCKED_MISSING / WIRE	TB_POS.BUS -> FK.IN / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W009 / BLOCKED_RATING / WIRE	K1.MAIN_NO -> FL.IN / MOTOR_SWITCHED	power_hc / Pack voltage when K1 closed / Red	Independent motor power removal	Prove E-stop drops K1, release alone does not re-latch, and stored bus energy decays
W010 / BLOCKED_RATING / WIRE	K1.MAIN_NO -> FR.IN / MOTOR_SWITCHED	power_hc / Pack voltage when K1 closed / Red	Independent motor power removal	Prove E-stop drops K1, release alone does not re-latch, and stored bus energy decays
W011 / BLOCKED_RATING / WIRE	K1.MAIN_NO -> FA.IN / MOTOR_SWITCHED	power_hc / Pack voltage when K1 closed / Red	Independent motor power removal	Prove E-stop drops K1, release alone does not re-latch, and stored bus energy decays
W012 / BLOCKED_RATING / WIRE	FL.OUT -> DRV_L.B+ / LEFT_SUPPLY	power_hc / Pack voltage / Red	FL; motor branch overload coordination	Verify sum of two left start/stall currents and actual IBT-2 thermal/connector capacity
W013 / BLOCKED_RATING / WIRE	FL.OUT -> C_BL.+ / LEFT_SUPPLY	power_hc / Pack voltage / Red	FL; motor branch overload coordination	Verify sum of two left start/stall currents and actual IBT-2 thermal/connector capacity
W014 / BLOCKED_RATING / WIRE	FL.OUT -> Z_L.+ / LEFT_SUPPLY	power_hc / Pack voltage / Red	FL; motor branch overload coordination	Verify sum of two left start/stall currents and actual IBT-2 thermal/connector capacity
W015 / BLOCKED_RATING / WIRE	FR.OUT -> DRV_R.B+ / RIGHT_SUPPLY	power_hc / Pack voltage / Red	FR; motor branch overload coordination	Verify sum of two right start/stall currents and actual IBT-2 thermal/connector capacity
W016 / BLOCKED_RATING / WIRE	FR.OUT -> C_BR.+ / RIGHT_SUPPLY	power_hc / Pack voltage / Red	FR; motor branch overload coordination	Verify sum of two right start/stall currents and actual IBT-2 thermal/connector capacity
W017 / BLOCKED_RATING / WIRE	FR.OUT -> Z_R.+ / RIGHT_SUPPLY	power_hc / Pack voltage / Red	FR; motor branch overload coordination	Verify sum of two right start/stall currents and actual IBT-2 thermal/connector capacity
W018 / BLOCKED_RATING / WIRE	FA.OUT -> DRV_A.Vs_12V / AUX_SUPPLY	power_hc / Pack voltage / Red	FA + FM5/FM6/FM7; package thermal review	Verify combined brush current, conveyor current, per-channel and shared thermal constraints
W019 / BLOCKED_RATING / WIRE	FA.OUT -> C_BA.+ / AUX_SUPPLY	power_hc / Pack voltage / Red	FA + FM5/FM6/FM7; package thermal review	Verify combined brush current, conveyor current, per-channel and shared thermal constraints
W020 / BLOCKED_RATING / WIRE	FA.OUT -> Z_A.+ / AUX_SUPPLY	power_hc / Pack voltage / Red	FA + FM5/FM6/FM7; package thermal review	Verify combined brush current, conveyor current, per-channel and shared thermal constraints

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W021 / BLOCKED_RATING / WIRE	F1.OUT -> BUCK1.IN+ / MCU_BUCK_IN	power_lc / Pack voltage / Red	F1 for buck input and lead	Verify module input range and fault-current/wire coordination
W022 / BLOCKED_RATING / WIRE	F2.OUT -> BUCK2.IN+ / LOGIC_BUCK_IN	power_lc / Pack voltage / Red	F2 for buck input and lead	Verify buck input range and module load budget
W023 / BLOCKED_MISSING / WIRE	OV1.OUT+ -> ESP1.5V_VIN / V5_MCU	power_lc / Regulated 5.0 V target / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	Measure output unloaded/loaded and verify actual DevKit regulated-5V input and USB backfeed path
W024 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> U1.20 (VCC) / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W025 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> DRV_L.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W026 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> DRV_R.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W027 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> DRV_A.5V_LOGIC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W028 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> US_F.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W029 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> US_L.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W030 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> US_R.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W031 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> C_U1.1 / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W032 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> C_F.1 / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W033 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> C_L.1 / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W034 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> C_R.1 / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W035 / PROVISIONAL / WIRE	ESP1.3V3 -> S2.SENSE_COM / V3V3	power_lc / ESP32 3.3 V / Red	MCU-derived 3V3; dry feedback and U2 only; never V5_LOGIC	Verify no connection to 5V/coil circuit and actual feedback-contact minimum current
W036 / PROVISIONAL / WIRE	ESP1.3V3 -> U2.8 (VCC) / V3V3	power_lc / ESP32 3.3 V / Red	MCU-derived 3V3; dry feedback and U2 only; never V5_LOGIC	Verify no connection to 5V/coil circuit and actual feedback-contact minimum current
W037 / PROVISIONAL / WIRE	ESP1.3V3 -> C_U2.1 / V3V3	power_lc / ESP32 3.3 V / Red	MCU-derived 3V3; dry feedback and U2 only; never V5_LOGIC	Verify no connection to 5V/coil circuit and actual feedback-contact minimum current
W038 / PROVISIONAL / WIRE	TB_GND.BUS -> BT1.PACK- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W039 / PROVISIONAL / WIRE	TB_GND.BUS -> P1.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W040 / PROVISIONAL / WIRE	BUCK1.OUT- -> ESP1.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W041 / PROVISIONAL / WIRE	TB_GND.BUS -> BUCK1.IN- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W042 / PROVISIONAL / INTERNAL_VERIFY	BUCK1.IN- -> BUCK1.OUT- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W043 / PROVISIONAL / WIRE	TB_GND.BUS -> BUCK2.IN- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W044 / PROVISIONAL / INTERNAL_VERIFY	BUCK2.IN- -> BUCK2.OUT- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W045 / PROVISIONAL / WIRE	TB_GND.BUS -> DRV_L.B- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W046 / PROVISIONAL / INTERNAL_VERIFY	DRV_L.B- -> DRV_L.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W047 / PROVISIONAL / WIRE	TB_GND.BUS -> DRV_R.B- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W048 / PROVISIONAL / INTERNAL_VERIFY	DRV_R.B- -> DRV_R.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W049 / PROVISIONAL / WIRE	TB_GND.BUS -> DRV_A.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W050 / PROVISIONAL / WIRE	BUCK2.OUT- -> US_F.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W051 / PROVISIONAL / WIRE	BUCK2.OUT- -> US_L.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W052 / PROVISIONAL / WIRE	BUCK2.OUT- -> US_R.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W053 / PROVISIONAL / WIRE	BUCK2.OUT- -> U1.10 (GND) / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W054 / PROVISIONAL / WIRE	U1.10 (GND) -> U1.1 (1OE_N) / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W055 / PROVISIONAL / WIRE	U1.10 (GND) -> U1.19 (2OE_N) / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W056 / PROVISIONAL / WIRE	TB_GND.BUS -> K1.A2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W057 / PROVISIONAL / WIRE	K1.A2 -> Z_COIL.A2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W058 / PROVISIONAL / WIRE	ESP1.GND -> R24.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W059 / PROVISIONAL / WIRE	ESP1.GND -> R26.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W060 / PROVISIONAL / WIRE	ESP1.GND -> R28.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W061 / PROVISIONAL / WIRE	U1.10 (GND) -> R1.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W062 / PROVISIONAL / WIRE	U1.10 (GND) -> R2.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W063 / PROVISIONAL / WIRE	U1.10 (GND) -> R3.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W064 / PROVISIONAL / WIRE	U1.10 (GND) -> R4.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W065 / PROVISIONAL / WIRE	U1.10 (GND) -> R5.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W066 / PROVISIONAL / WIRE	U1.10 (GND) -> R6.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W067 / PROVISIONAL / WIRE	U1.10 (GND) -> R7.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W068 / PROVISIONAL / WIRE	U1.10 (GND) -> R8.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W069 / PROVISIONAL / WIRE	DRV_A.GND -> R9.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W070 / PROVISIONAL / WIRE	DRV_A.GND -> R10.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W071 / PROVISIONAL / WIRE	DRV_A.GND -> R11.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W072 / PROVISIONAL / WIRE	DRV_A.GND -> R12.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W073 / PROVISIONAL / WIRE	DRV_L.GND -> R13.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W074 / PROVISIONAL / WIRE	DRV_L.GND -> R14.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W075 / PROVISIONAL / WIRE	DRV_R.GND -> R15.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W076 / PROVISIONAL / WIRE	DRV_R.GND -> R16.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W077 / PROVISIONAL / WIRE	DRV_A.GND -> R17.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W078 / PROVISIONAL / WIRE	DRV_A.GND -> R18.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W079 / PROVISIONAL / WIRE	DRV_L.GND -> R19.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W080 / PROVISIONAL / WIRE	DRV_L.GND -> R20.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W081 / PROVISIONAL / WIRE	DRV_R.GND -> R21.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W082 / PROVISIONAL / WIRE	DRV_R.GND -> R22.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W083 / PROVISIONAL / WIRE	ESP1.GND -> R29.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W084 / PROVISIONAL / WIRE	U1.10 (GND) -> C_U1.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W085 / PROVISIONAL / WIRE	US_F.GND -> C_F.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W086 / PROVISIONAL / WIRE	US_L.GND -> C_L.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W087 / PROVISIONAL / WIRE	US_R.GND -> C_R.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W088 / PROVISIONAL / WIRE	DRV_L.B- -> C_BL.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W089 / PROVISIONAL / WIRE	DRV_R.B- -> C_BR.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W090 / PROVISIONAL / WIRE	DRV_A.GND -> C_BA.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W091 / PROVISIONAL / WIRE	BUCK1.OUT- -> OV1.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W092 / PROVISIONAL / WIRE	BUCK2.OUT- -> OV2.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W093 / PROVISIONAL / WIRE	DRV_L.B- -> Z_L.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W094 / PROVISIONAL / WIRE	DRV_R.B- -> Z_R.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W095 / PROVISIONAL / WIRE	DRV_A.GND -> Z_A.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W096 / PROVISIONAL / WIRE	ESP1.GND -> U2.4 (GND) / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W097 / PROVISIONAL / WIRE	U2.4 (GND) -> C_U2.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W098 / PROVISIONAL / WIRE	ESP1.GND -> R30.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W099 / PROVISIONAL / WIRE	ESP1.GND -> R31.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W100 / PROVISIONAL / WIRE	ESP1.GND -> R32.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W101 / BLOCKED_RATING / WIRE	BT1.BAL_CELL0 -> P1.CELL0 / CELL0	sensor / Cell-stack potential; order TBD / Green	Sense lead protection and connector order per selected P1	No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips
W102 / BLOCKED_RATING / WIRE	BT1.BAL_CELL1 -> P1.CELL1 / CELL1	sensor / Cell-stack potential; order TBD / Green	Sense lead protection and connector order per selected P1	No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips
W103 / BLOCKED_RATING / WIRE	BT1.BAL_CELL2 -> P1.CELL2 / CELL2	sensor / Cell-stack potential; order TBD / Green	Sense lead protection and connector order per selected P1	No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips
W104 / BLOCKED_RATING / WIRE	BT1.BAL_CELL3 -> P1.CELL3 / CELL3	sensor / Cell-stack potential; order TBD / Green	Sense lead protection and connector order per selected P1	No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W105 / BLOCKED_RATING / WIRE	FK.OUT -> OV1.PG_COM / COIL_FUSED	power_lc / Pack voltage; K1 coil compatibility TBD / Red	FK + S2 coil NC	Coil must operate across actual loaded pack range, or a qualified coil supply is required
W106 / BLOCKED_MISSING / WIRE	S2.COIL_NC -> S3.COM / COIL_STOP_CHAIN	power_lc / Coil supply when stop released / Red	Independent NC E-stop chain	Pressing stop must break coil circuit even if MCU hung
W107 / BLOCKED_MISSING / WIRE	S2.COIL_NC -> K1.AUX_HOLD_COM / COIL_STOP_CHAIN	power_lc / Coil supply when stop released / Red	Independent NC E-stop chain	Pressing stop must break coil circuit even if MCU hung
W108 / BLOCKED_MISSING / WIRE	S3.NO -> K1.AUX_HOLD_NO / COIL_HOLD	power_lc / Coil supply when manually latched / Red	Momentary S3 in parallel with K1 NO auxiliary	Verify seal-in, dropout, no automatic restart on stop release/power restoration, and clamp release time
W109 / BLOCKED_MISSING / WIRE	S3.NO -> K1.A1 / COIL_HOLD	power_lc / Coil supply when manually latched / Red	Momentary S3 in parallel with K1 NO auxiliary	Verify seal-in, dropout, no automatic restart on stop release/power restoration, and clamp release time
W110 / BLOCKED_MISSING / WIRE	S3.NO -> Z_COIL.A1 / COIL_HOLD	power_lc / Coil supply when manually latched / Red	Momentary S3 in parallel with K1 NO auxiliary	Verify seal-in, dropout, no automatic restart on stop release/power restoration, and clamp release time
W111 / BLOCKED_MISSING / WIRE	K1.AUX_SENSE_NO -> ESP1.GPIO27 / ESTOP_OK	control / Healthy 3.3 V; open/pressed 0 V / Blue	Isolated NC contact, R29 pull-down; firmware internal pull-down	HIGH only when S2 released AND K1 closed. Open contact/wire or power-good loss trips; initial K1-open boot latches ESTOP until manual power-arm, RESET, ARM.
W112 / BLOCKED_MISSING / WIRE	K1.AUX_SENSE_NO -> R29.1 / ESTOP_OK	control / Healthy 3.3 V; open/pressed 0 V / Blue	Isolated NC contact, R29 pull-down; firmware internal pull-down	HIGH only when S2 released AND K1 closed. Open contact/wire or power-good loss trips; initial K1-open boot latches ESTOP until manual power-arm, RESET, ARM.
W113 / BLOCKED_MISSING / WIRE	ESP1.GPIO16 -> U1.2 (1A1) / L_FWD_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R1	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W114 / BLOCKED_MISSING / WIRE	ESP1.GPIO16 -> R1.1 / L_FWD_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R1	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W115 / BLOCKED_MISSING / WIRE	U1.18 (1Y1) -> DRV_L.RPWM / L_FWD_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W116 / BLOCKED_MISSING / WIRE	U1.18 (1Y1) -> R19.1 / L_FWD_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W117 / BLOCKED_MISSING / WIRE	ESP1.GPIO17 -> U1.4 (1A2) / L_REV_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R2	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W118 / BLOCKED_MISSING / WIRE	ESP1.GPIO17 -> R2.1 / L_REV_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R2	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W119 / BLOCKED_MISSING / WIRE	U1.16 (1Y2) -> DRV_L.LPWM / L_REV_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W120 / BLOCKED_MISSING / WIRE	U1.16 (1Y2) -> R20.1 / L_REV_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W121 / BLOCKED_MISSING / WIRE	ESP1.GPIO18 -> U1.6 (1A3) / R_FWD_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R3	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W122 / BLOCKED_MISSING / WIRE	ESP1.GPIO18 -> R3.1 / R_FWD_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R3	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W123 / BLOCKED_MISSING / WIRE	U1.14 (1Y3) -> DRV_R.RPWM / R_FWD_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W124 / BLOCKED_MISSING / WIRE	U1.14 (1Y3) -> R21.1 / R_FWD_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W125 / BLOCKED_MISSING / WIRE	ESP1.GPIO19 -> U1.8 (1A4) / R_REV_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R4	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W126 / BLOCKED_MISSING / WIRE	ESP1.GPIO19 -> R4.1 / R_REV_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R4	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W127 / BLOCKED_MISSING / WIRE	U1.12 (1Y4) -> DRV_R.LPWM / R_REV_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W128 / BLOCKED_MISSING / WIRE	U1.12 (1Y4) -> R22.1 / R_REV_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W129 / BLOCKED_MISSING / WIRE	ESP1.GPIO14 -> U1.11 (2A1) / ENABLE_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R5	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W130 / BLOCKED_MISSING / WIRE	ESP1.GPIO14 -> R5.1 / ENABLE_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R5	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W131 / BLOCKED_MISSING / WIRE	U1.9 (2Y1) -> S2.EN_COM / ENABLE_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W132 / BLOCKED_MISSING / WIRE	ESP1.GPIO32 -> U1.13 (2A2) / TF_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R6	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W133 / BLOCKED_MISSING / WIRE	ESP1.GPIO32 -> R6.1 / TF_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R6	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W134 / BLOCKED_MISSING / WIRE	U1.7 (2Y2) -> US_F.TRIG / TF_5V	sensor / 0/5 V / Green	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W135 / BLOCKED_MISSING / WIRE	ESP1.GPIO33 -> U1.15 (2A3) / TL_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R7	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W136 / BLOCKED_MISSING / WIRE	ESP1.GPIO33 -> R7.1 / TL_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R7	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W137 / BLOCKED_MISSING / WIRE	U1.5 (2Y3) -> US_L.TRIG / TL_5V	sensor / 0/5 V / Green	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W138 / BLOCKED_MISSING / WIRE	ESP1.GPIO13 -> U1.17 (2A4) / TR_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R8	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W139 / BLOCKED_MISSING / WIRE	ESP1.GPIO13 -> R8.1 / TR_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R8	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W140 / BLOCKED_MISSING / WIRE	U1.3 (2Y4) -> US_R.TRIG / TR_5V	sensor / 0/5 V / Green	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W141 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_L.R_EN / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W142 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_L.L_EN / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W143 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_R.R_EN / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W144 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_R.L_EN / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W145 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_A.ENA / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W146 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_A.ENB / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W147 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R13.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W148 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R14.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W149 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R15.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W150 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R16.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W151 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R17.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W152 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R18.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W153 / PROVISIONAL / WIRE	ESP1.GPIO21 -> DRV_A.IN1 / BRUSH_IN1	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W154 / PROVISIONAL / WIRE	ESP1.GPIO21 -> R9.1 / BRUSH_IN1	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W155 / PROVISIONAL / WIRE	ESP1.GPIO22 -> DRV_A.IN2 / BRUSH_IN2	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W156 / PROVISIONAL / WIRE	ESP1.GPIO22 -> R10.1 / BRUSH_IN2	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W157 / PROVISIONAL / WIRE	ESP1.GPIO23 -> DRV_A.IN3 / CONVEYOR_IN3	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W158 / PROVISIONAL / WIRE	ESP1.GPIO23 -> R11.1 / CONVEYOR_IN3	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W159 / PROVISIONAL / WIRE	ESP1.GPIO25 -> DRV_A.IN4 / CONVEYOR_IN4	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W160 / PROVISIONAL / WIRE	ESP1.GPIO25 -> R12.1 / CONVEYOR_IN4	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W161 / BLOCKED_MISSING / WIRE	US_F.ECHO -> R23.1 / EF_RAW	sensor / Standard module: potentially 0/5 V / Green	2.2k upper / 3.3k lower 1% divider required	Keep actual ECHO disconnected until parts fitted and high/low/overshoot verified independently of ESP32
W162 / BLOCKED_MISSING / WIRE	R23.2 -> R24.1 / EF_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W163 / BLOCKED_MISSING / WIRE	R23.2 -> U2.1 (1A) / EF_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W164 / BLOCKED_MISSING / WIRE	U2.7 (1Y) -> ESP1.GPIO34 / EF_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W165 / BLOCKED_MISSING / WIRE	U2.7 (1Y) -> R30.1 / EF_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W166 / BLOCKED_MISSING / WIRE	US_L.ECHO -> R25.1 / EL_RAW	sensor / Standard module: potentially 0/5 V / Green	2.2k upper / 3.3k lower 1% divider required	Keep actual ECHO disconnected until parts fitted and high/low/overshoot verified independently of ESP32
W167 / BLOCKED_MISSING / WIRE	R25.2 -> R26.1 / EL_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W168 / BLOCKED_MISSING / WIRE	R25.2 -> U2.3 (2A) / EL_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W169 / BLOCKED_MISSING / WIRE	U2.5 (2Y) -> ESP1.GPIO35 / EL_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W170 / BLOCKED_MISSING / WIRE	U2.5 (2Y) -> R31.1 / EL_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W171 / BLOCKED_MISSING / WIRE	US_R.ECHO -> R27.1 / ER_RAW	sensor / Standard module: potentially 0/5 V / Green	2.2k upper / 3.3k lower 1% divider required	Keep actual ECHO disconnected until parts fitted and high/low/overshoot verified independently of ESP32
W172 / BLOCKED_MISSING / WIRE	R27.2 -> R28.1 / ER_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W173 / BLOCKED_MISSING / WIRE	R27.2 -> U2.6 (3A) / ER_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W174 / BLOCKED_MISSING / WIRE	U2.2 (3Y) -> ESP1.GPIO36 / ER_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W175 / BLOCKED_MISSING / WIRE	U2.2 (3Y) -> R32.1 / ER_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W176 / BLOCKED_RATING / WIRE	DRV_L.M+ -> FM1.IN / DRV_L_MP	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W177 / BLOCKED_RATING / WIRE	DRV_L.M+ -> FM2.IN / DRV_L_MP	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W178 / BLOCKED_RATING / WIRE	DRV_L.M- -> M1.B / DRV_L_MN	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W179 / BLOCKED_RATING / WIRE	DRV_L.M- -> M2.B / DRV_L_MN	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W180 / BLOCKED_RATING / WIRE	FM1.OUT -> M1.A / M1_A	motor / Bridge output after fuse / Orange	FM1; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W181 / BLOCKED_RATING / WIRE	FM2.OUT -> M2.A / M2_A	motor / Bridge output after fuse / Orange	FM2; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W182 / BLOCKED_RATING / WIRE	DRV_R.M+ -> FM3.IN / DRV_R_MP	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W183 / BLOCKED_RATING / WIRE	DRV_R.M+ -> FM4.IN / DRV_R_MP	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W184 / BLOCKED_RATING / WIRE	DRV_R.M- -> M3.B / DRV_R_MN	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W185 / BLOCKED_RATING / WIRE	DRV_R.M- -> M4.B / DRV_R_MN	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W186 / BLOCKED_RATING / WIRE	FM3.OUT -> M3.A / M3_A	motor / Bridge output after fuse / Orange	FM3; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W187 / BLOCKED_RATING / WIRE	FM4.OUT -> M4.A / M4_A	motor / Bridge output after fuse / Orange	FM4; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W188 / BLOCKED_RATING / WIRE	DRV_A.OUT1 -> FM5.IN / DRV_A_OUT1	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W189 / BLOCKED_RATING / WIRE	DRV_A.OUT1 -> FM6.IN / DRV_A_OUT1	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W190 / BLOCKED_RATING / WIRE	DRV_A.OUT2 -> M5.B / DRV_A_OUT2	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W191 / BLOCKED_RATING / WIRE	DRV_A.OUT2 -> M6.B / DRV_A_OUT2	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W192 / BLOCKED_RATING / WIRE	FM5.OUT -> M5.A / M5_A	motor / Bridge output after fuse / Orange	FM5; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W193 / BLOCKED_RATING / WIRE	FM6.OUT -> M6.A / M6_A	motor / Bridge output after fuse / Orange	FM6; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W194 / BLOCKED_RATING / WIRE	DRV_A.OUT3 -> FM7.IN / DRV_A_OUT3	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W195 / BLOCKED_RATING / WIRE	DRV_A.OUT4 -> M7.B / DRV_A_OUT4	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W196 / BLOCKED_RATING / WIRE	FM7.OUT -> M7.A / M7_A	motor / Bridge output after fuse / Orange	FM7; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W197 / BLOCKED_MISSING / WIRE	BUCK1.OUT+ -> OV1.IN+ / V5_MCU_RAW	power_lc / 5V target; buck-fault voltage may approach pack / Red	OV1 required; never assume LM2596 is an overvoltage protector	Select actual independent output protection and supervisor; simulate abnormal voltage with electronics disconnected
W198 / BLOCKED_MISSING / WIRE	BUCK2.OUT+ -> OV2.IN+ / V5_LOGIC_RAW	power_lc / 5V target; buck-fault voltage may approach pack / Red	OV2 required; never assume LM2596 is an overvoltage protector	Select actual independent output protection and supervisor; simulate abnormal voltage with electronics disconnected
W199 / BLOCKED_MISSING / WIRE	OV1.PG_NO -> OV2.PG_COM / COIL_PG1	power_lc / Coil supply; contact isolation required / Red	Both logic rails must be healthy before K1 can energize/hold	A missing/abnormal rail opens the coil chain; restoration alone must not re-latch K1. Select actual supervisor/interface circuitry.
W200 / BLOCKED_MISSING / WIRE	OV2.PG_NO -> S2.COIL_COM / COIL_PG2	power_lc / Coil supply; contact isolation required / Red	Both logic rails must be healthy before K1 can energize/hold	A missing/abnormal rail opens the coil chain; restoration alone must not re-latch K1. Select actual supervisor/interface circuitry.
W201 / BLOCKED_MISSING / WIRE	S2.SENSE_NC -> K1.AUX_SENSE_COM / SENSE_STOP_OK	control / 3.3 V only / Blue	Separate dry K1 NO feedback contact in series with S2 NC	Map actual contact terminals; keep isolated from seal-in coil circuit. No jumper around this interlock.